Report Gasoline

Time Series - MESIO UPC-UB

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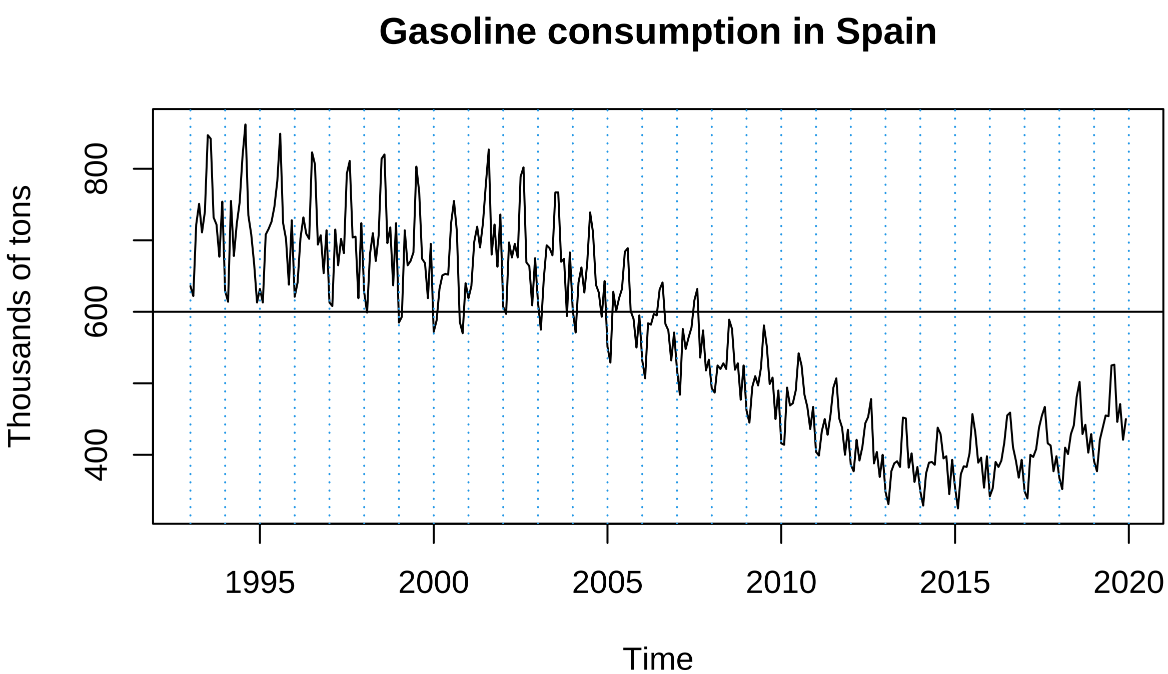
Classic ARIMA approach

Model identification



The model has a mean of 563.19 and a variance of 136.34. Since we detect this time serie is not stationary, we applied the next transformations:

- Box-Cox transformation:
- Seasonality:
- Non-constant variance:

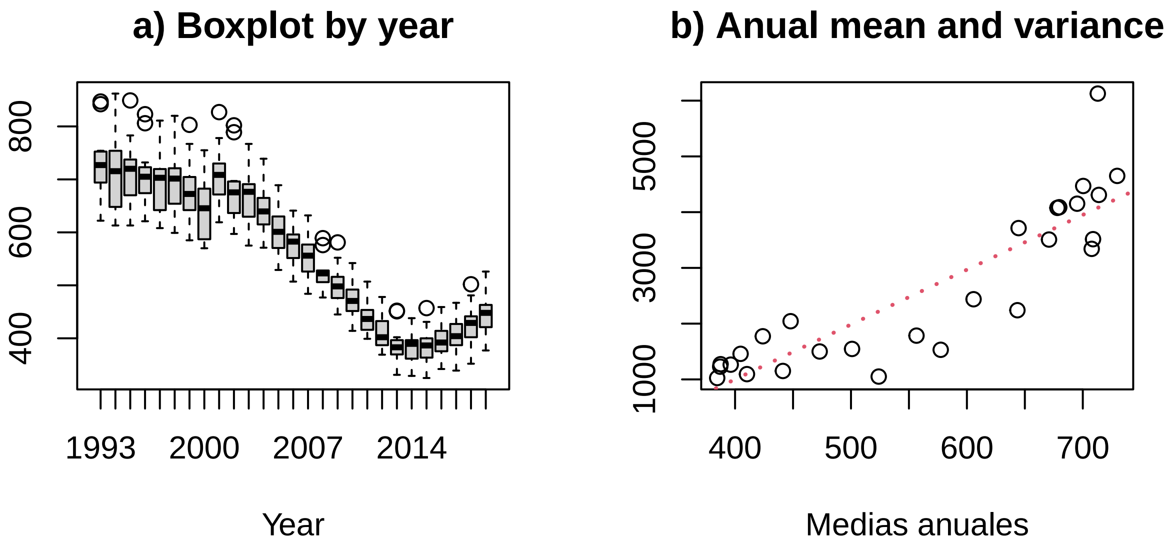
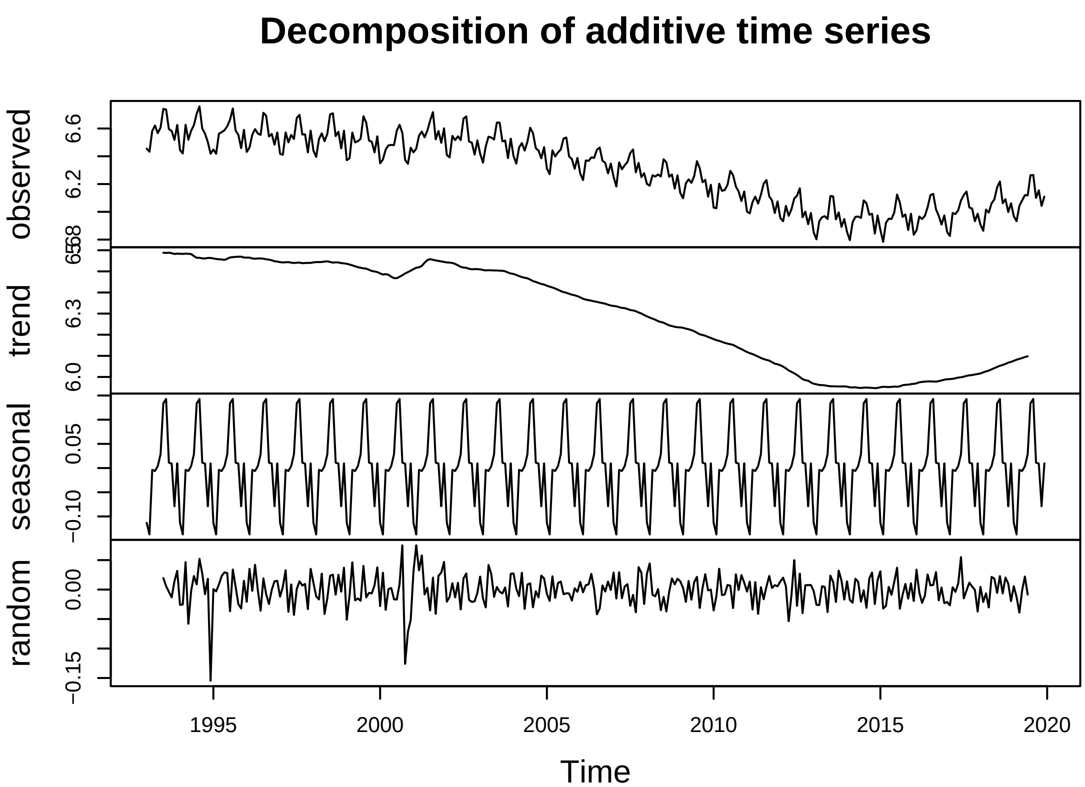
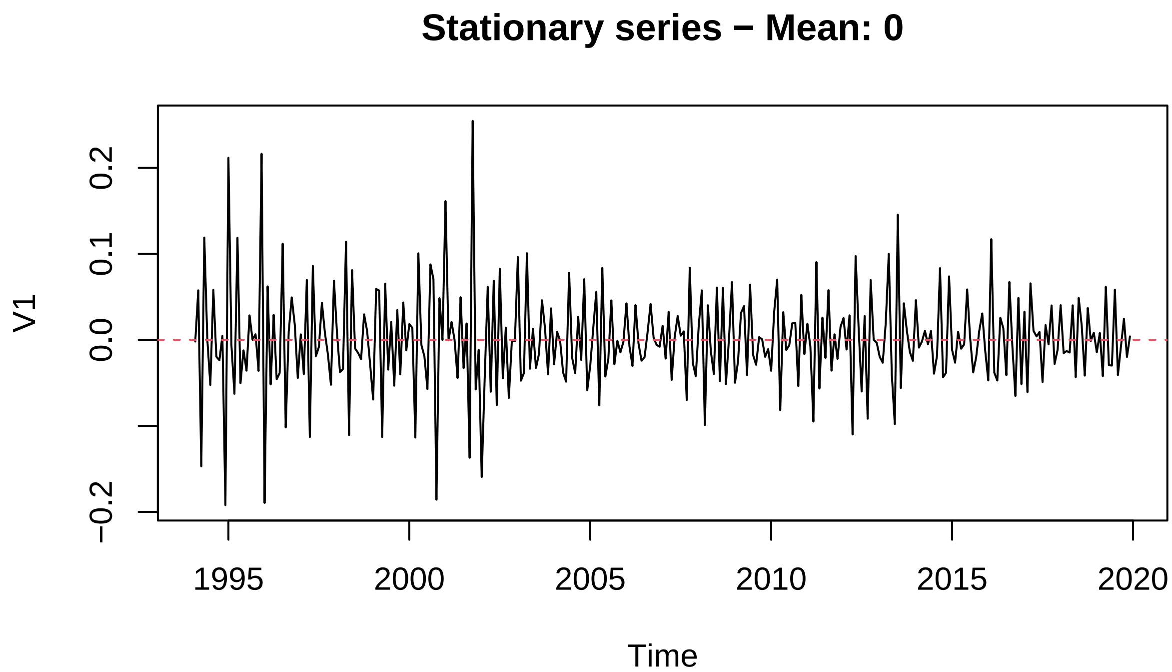


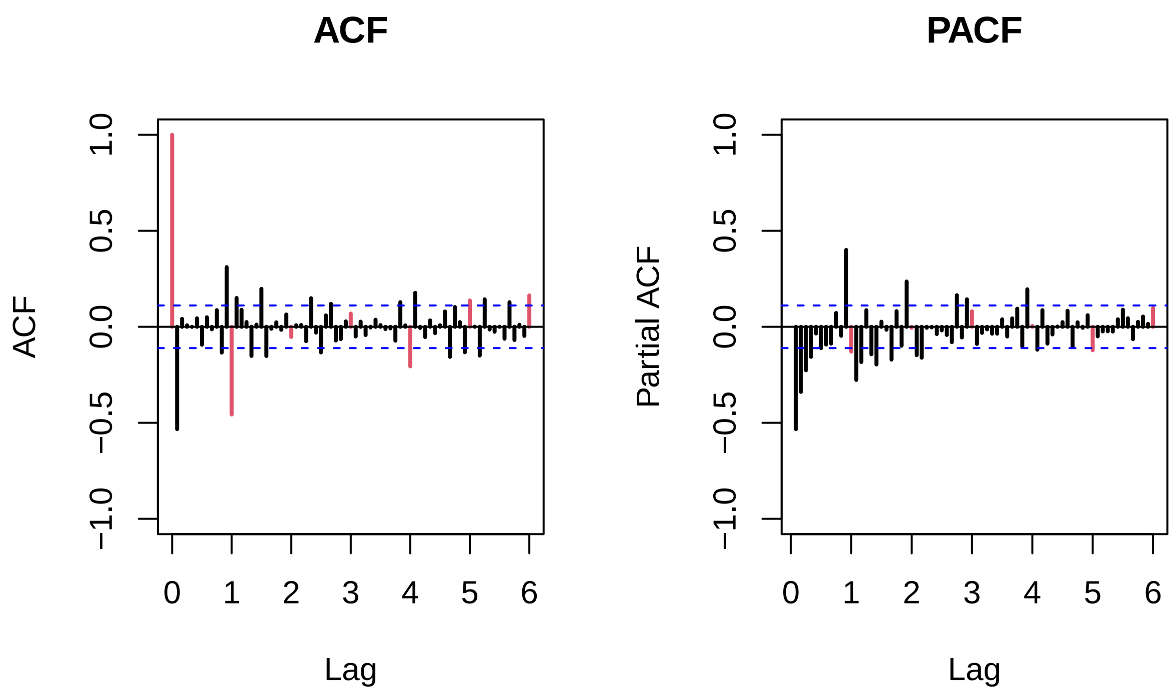
Figure 1: Non-constant variance serie





Estimation for plausible models

In the model identification phase of the Box-Jenkins methodology, we analyze the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots of the suitably differenced stationary time series to determine the appropriate autoregressive (AR) and moving average (MA) terms. Based on the visual inspection of the provided correlograms, we divide our identification process into two parts: the seasonal component and the regular (non-seasonal) component.



Seasonal Component Identification: Because PACF shows significant activity at the first seasonal lag—rather than a clear, sharp cutoff that is shown in the ACF at the first seasonal lag moreover, a mixed seasonal process is the most appropriate choice. Therefore, we select a Seasonal ARMA(1,1) model to capture both the seasonal autoregressive (SAR) and seasonal moving average (SMA).

Regular Component Identification:

- MA(1): Proposed under the assumption that the regular ACF effectively cuts off after the very first lag, while the PACF tails off
- ARMA(1,1): assuming both the ACF and PACF exhibit a tapering/decaying pattern from the first lag, capturing both a short-term autoregressive and moving average effect.
- MA(2): Proposed to account for the possibility that the first two lags in the ACF are driving the moving average process before cutting off
- ARMA(1,2) : mixed model included to capture any residual moving average dependencies up to the second lag while maintaining a first-order autoregressive term.

Table 1: Model Comparison

Models	Log-Likelihood	AIC	AR(1)	MA(1)	MA(2)	SMA(1)
MA(1)	610.97	-1215.93	NA	-0.68	NA	-0.89
ARMA(1,1)	613.46	-1218.93	-0.17	-0.60	NA	-0.88
MA(2)	**613.95**	**-1219.89**	**NA**	**-0.79**	**0.14**	**-0.88**
ARMA(1,2)	**617.75**	**-1225.50**	**0.99**	**-1.73**	**0.75**	**-0.92**

As shown in the table above, an estimation comparison was conducted between the 4 selected models based on their goodness of fit and complexity. The results of **log likelihood** and **AIC** values indicate that **MA(2)** and **ARMA(1,2)** exhibit the two best overall performing models.

Both models' residuals were evaluated, reaching the normality and independence requirements, as can be see in the next figures. However, the Ljung-Box test shows significant p-values starting lag 12 approximately, which is problematic since it means certain degree of autocorrelation in residuals[1].

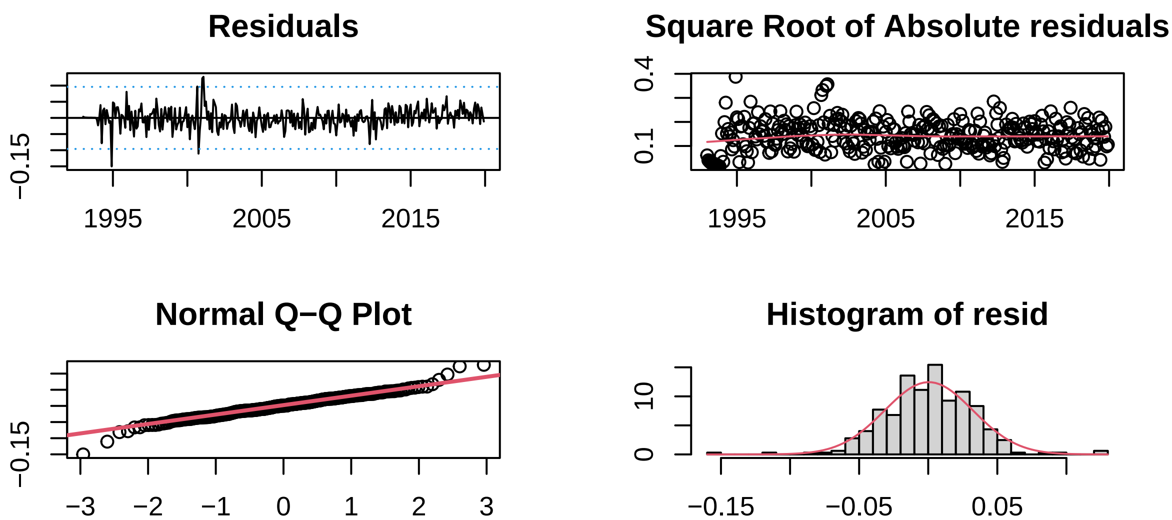


Figure 2: MA(2) residual validation

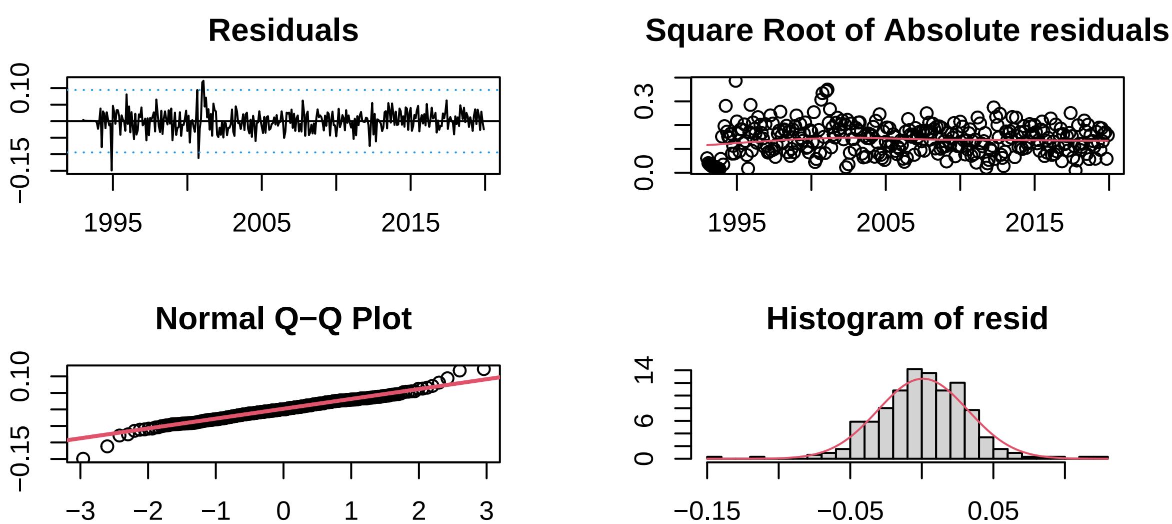


Figure 3: ARMA(1,2) residual validation

Prediction assesment and forecasting

To evaluate the prediction performance, we checked the models' stability, it means the comparison between the coefficients of the model with the full series and with a restricted series without the las 12 observations. Both MA(2) and ARMA(1,2) showed stability since their coefficients don't change significantly under those conditions.

In terms of prediction, the plots show how MA(2) and ARMA(1,2) perform trying to forecast the values for

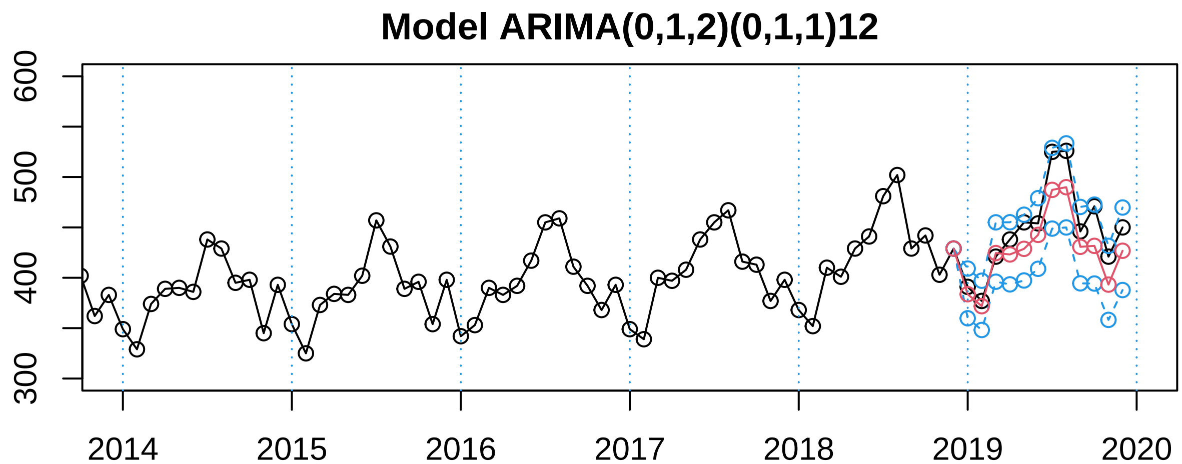
[1]See details in Appendices

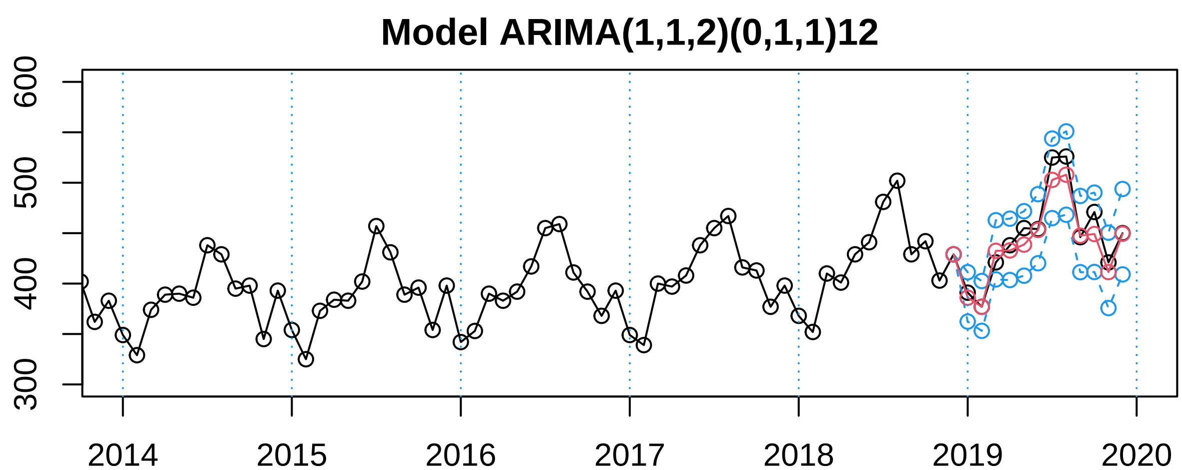
Table 2: MA(2)(left) and ARMA(1,2)(right) stability

Variables	Full	Restricted	Diference
ma1	-0.79	-0.78	0.01
ma2	0.14	0.13	0.01
sma1	-0.88	-0.89	0.01

Variables	Full	Restricted	Diference
ar1	0.99	0.99	0.00
ma1	-1.73	-1.72	0.01
ma2	0.75	0.74	0.01
sma1	-0.92	-0.93	0.01

the last year. Visualy speaking, we can see ARMA(1,2) has predicted points (in red) closer to the real value points (in black).





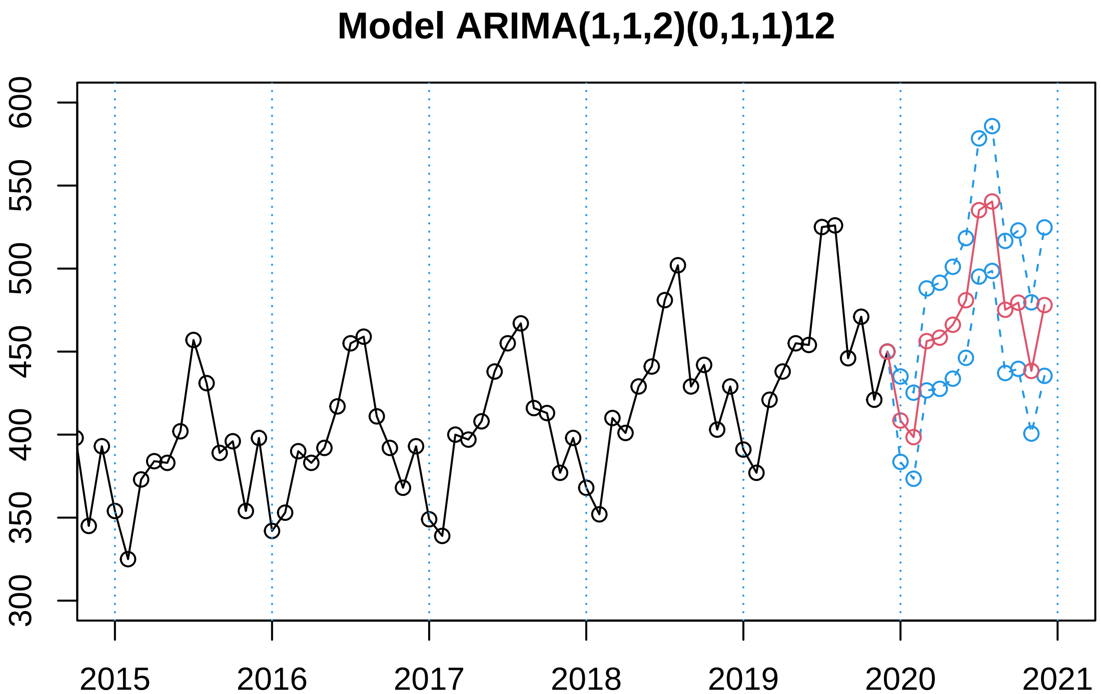
The metrics also show better performance for ARMA(1,2) mainly for RMSE (12.4) and MAE (9.39). Both models reach a mean confidence interval close to 63, and very similar RMSPE and MAPE.

Model	RMSE	MAE	RMSPE	MAPE	Mean CI
ARIMA(0,1,2)	24.06	20.66	0.05	0.04	63.65
ARiMA(1,1,2)	12.40	9.39	0.03	0.02	63.52

At this point, our best model by Information Criteria and prediction performance seems to be the SARIMA(1,1,2)x(0,1,1), which can be formally described as:

$$(1-\phi_1 B)(1-B)(1-B^{12})X_t = (1+\theta_1 B + \theta_2 B^2)(1+\Theta_1 B^{12})Z_t$$

With this full model, the forecast for the next 12 months is described in the next plot. As can be expected, the prediction follows the seasonal pattern very well, imitating the anual shape of the curve.



However, the selected model still have a big problem: auto-regresive residuals. As we mentioned before, none of the proposed models passed the Ljung-Box test, which means that there are certain degree of auto-correlation in the residuals of the models. Since none of the proposed model avoid the problem we follow the next hypothesis: this could be corrected by taking into account some calendar effects and with automatic outlier detection.

ARIMAX Analysis: Outlier and Calendar Effects Treatment

Following our hypothesis, and according to the nature of gasoline consumption, we belive there are two main calendar effects that are affecting our series: labor days and Easter holidays. It follows because on labor days there is more transit due to labor and educational activities. This pushes up the gasoline consumption and with more labor days in a month the consumption will rise. Easter holidays are a special case since it implies a high level of tourism and travels, but also because it changes its dates every year between March and April. So, we should take it into account these two conditions as important calendar effects.

To test the relevance of calendar effects we are creating two dummy variables to identify labor days and Easter across our series. Next, we use those as exogenous variables in our best model (in this case, ARIMA(1,1,2)). Then we test the significance of the model's terms as the ratio between coefficients and standard errors.

```
##        ar1        ma1        ma2       sma1  wTradDays      wEast
##  66.242567 -39.617071  17.128261 -21.522962   8.108000   3.943221
```

Table 3: Calendar Effect: coefficient validation

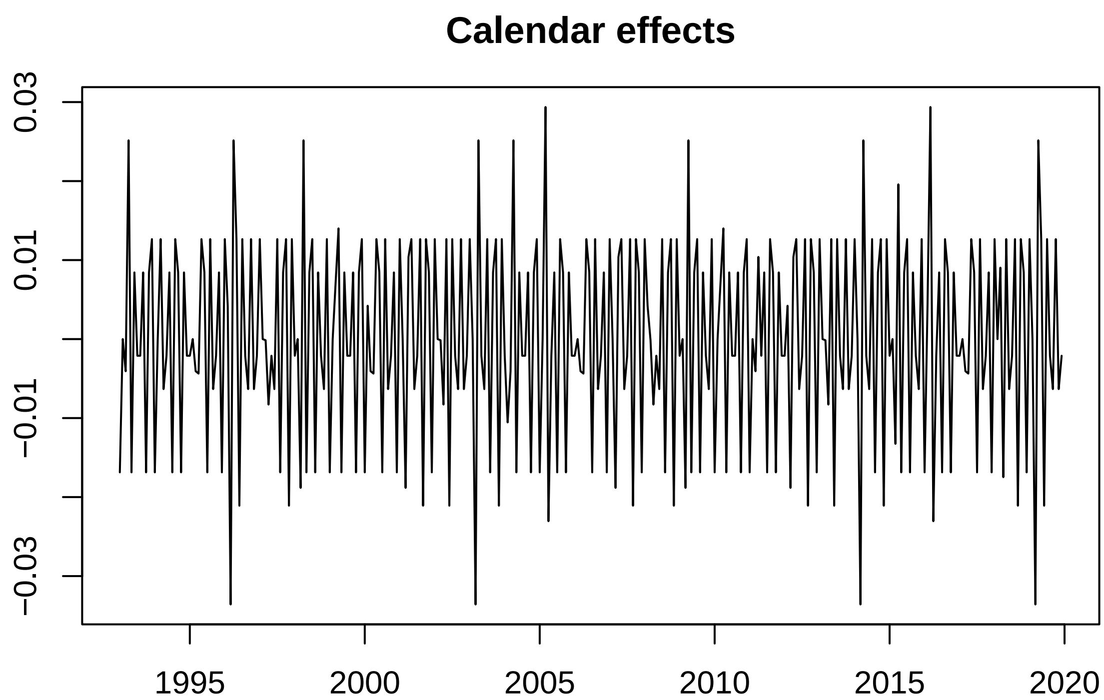
term	estimate	std.error	Significance
ar1	0.99	0.01	66.24
ma1	-1.68	0.04	-39.62
ma2	0.70	0.04	17.13
sma1	-0.90	0.04	-21.52
wTradDays	0.00	0.00	8.11
wEast	0.03	0.01	3.94

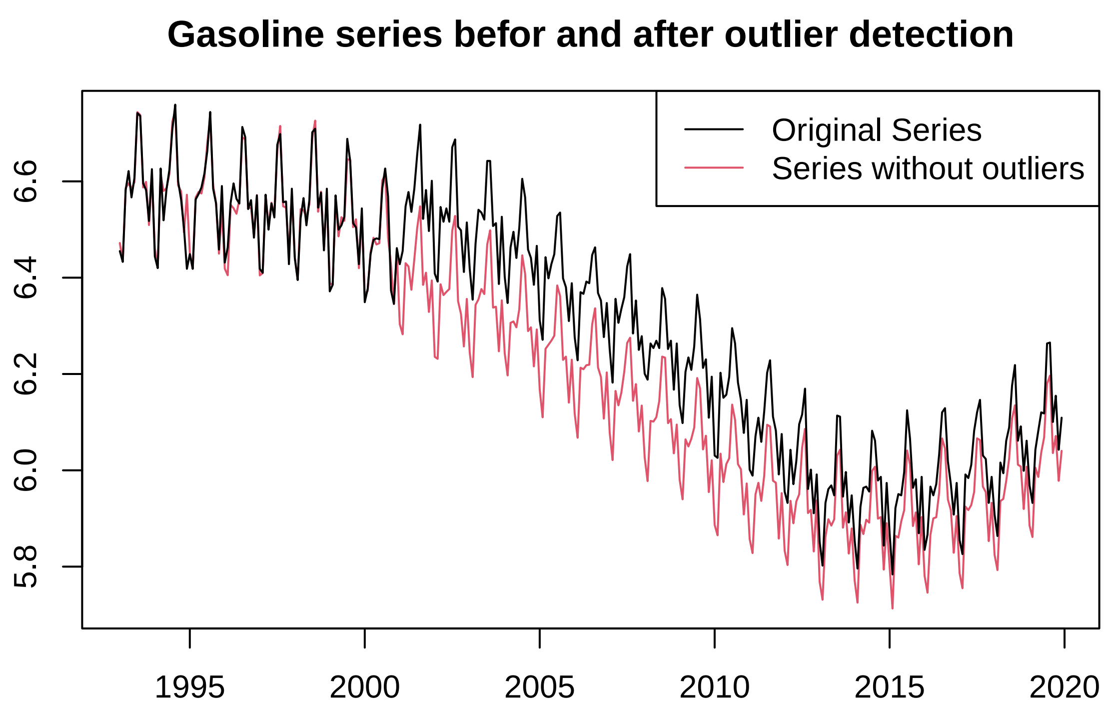
As can be seen in the table, the exogenous variables show to be significant to explain the observed series. After this, we are using the `outdetec` function to detect different kind of outliers: additive outliers, transitive change and level shift. The next table shows the observations identified as outlier and their classification.

Table 4: Detected outliers in Gasoline consumption series

	Obs	type_detected	W_coeff	ABS_L_Ratio	Date	perc.Obs
10	16	AO	-0.055	2.968	Abr 1994	-0.055
2	24	AO	-0.152	6.876	Dic 1994	-0.152
7	36	AO	0.064	3.313	Dic 1995	0.064
9	38	TC	0.054	3.171	Feb 1996	0.054
19	79	AO	0.045	2.674	Jul 1999	0.045
3	93	LS	0.092	5.596	Sep 2000	0.092
1	94	TC	-0.143	6.650	Oct 2000	-0.143
4	97	LS	0.069	4.325	Ene 2001	0.069
15	98	AO	0.046	2.641	Feb 2001	0.046
8	108	AO	0.064	3.366	Dic 2001	0.064
14	164	AO	-0.047	2.662	Ago 2006	-0.047
11	171	AO	0.049	2.694	Mar 2007	0.049
18	182	AO	0.046	2.691	Feb 2008	0.046
12	186	TC	-0.045	2.723	Jun 2008	-0.045
16	219	LS	-0.036	2.681	Mar 2011	-0.036
5	232	LS	-0.054	3.428	Abr 2012	-0.054
6	234	AO	0.081	4.106	Jun 2012	0.081
17	278	AO	0.046	2.631	Feb 2016	0.046
13	294	AO	0.048	2.632	Jun 2017	0.048

At this point, with calendar effects and outlier detection, we can "clean" our series taking out those atypical patterns. The new series will be a linear combination of the original one and its outliers. Then, we will fit our selected model over this new series, with calendar effects as regressors to extract their coefficients. With estimated effects, our clean series would result from the free-of-outlier series minus their calendar effects. Here we can see the estimated calendar effects and our original series along with the clean one.





ARIMAX model identification

Once we have our new series, we repeat the identification process from the ACF and PACF plots. For this series we are considering again one regular and one seasonal differentiation to keep it stationary.

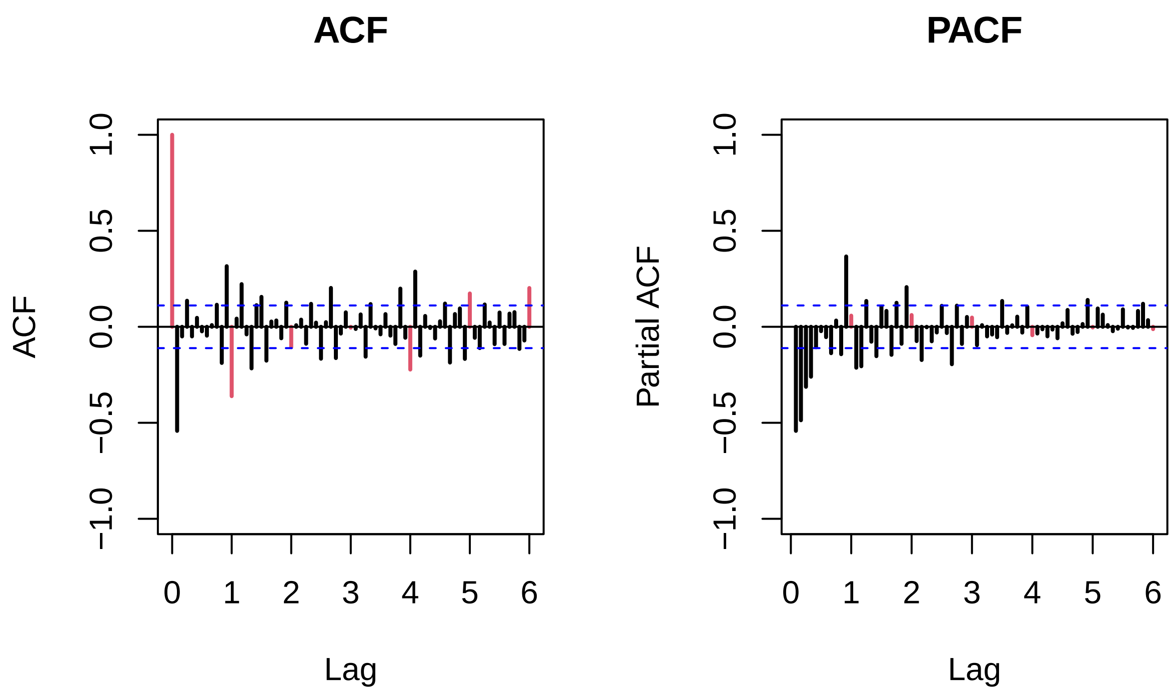


Figure 4: ARIMAX model identification

Proposed models for clean series: Regular MA(1), MA(3), MA(11) and ARIMA(1,2) with Seasonal MA(1).

The first insight we extract from the plots is the clear seasonal pattern, which can be described as an moving average of level 1 (MA(1)). For the regular part we identify this possible combinations: MA(1), MA(3), MA(11), ARMA(1,5) and ARMA(1,2)[2]. After fitting and testing the candidates, we compare them by Log-likelihood and AIC.

Table 5: ARIMAX Models Comparison

ARIMAX Model	Log-Likelihood	AIC
MA(1)	775.46	-1540.93
MA(2)	777.30	-1542.61
MA(11)	**795.44**	**-1560.89**
ARMA(1,2)	**789.87**	**-1565.74**
ARMA(1,5)	782.57	-1545.13

In general terms, the five models reach an important goal: they improve the AIC of ARIMA models by far, from around -1200 to -1500. The two best performing models were MA(11) and ARMA(1,2). The first one is an model adjusted using "brute force", which works but makes the model much more complex. On the other hand, ARMA(1,2) keeps the structure identified in the ARIMA approach, and here performs very well reaching the best AIC. Also both of them achieve all validation requirements, specially the Ljung-Box test that failed for the ARIMA models. For these reasons, we will test their performance in prediction, using the last year as the test sample.

[2]ARMA(1,2) was tested because it was our selected model on ARIMA approach

Table 6: ARIMAX Models Prediction assessment

Model	RMSE	MAE	RMSPE	MAPE	Mean CI
ARIMAX(0,1,11)	34.71	35.37	0.08	0.08	34.63
ARIMAX(1,1,2)	30.36	30.72	0.07	0.07	35.36

ARIMA and ARIMAX models comparison

Finally, we proceed to compare the two ARIMA and the two ARIMAX models. Here are our results.

Table 7: ARIMA and ARIMAX models comparison

Models	Log-Likelihood	AIC	RMSE	MAE	RMSPE	MAPE	Mean CI
ARIMA(0,1,2)	613.95	-1219.89	24.06	20.66	0.05	0.04	63.65
ARiMA(1,1,2)	617.75	-1225.50	12.40	9.39	0.03	0.02	63.52
ARIMAX(0,1,11)	795.44	-1560.89	34.71	35.37	0.08	0.08	34.63
ARIMAX(1,1,2)	**789.87**	**-1565.74**	**30.36**	**30.72**	**0.07**	**0.07**	**35.36**

Looking the results, we realized there is a paradox. On one side we have ARIMA models with great point prediction metrics. We could think this is perfect for forecast and future decisions about the gasoline. But they don't take into account the outilers and calendar effects. On the other side, ARIMAX models don't perform as well in RMSE, MAE and other metrics. Instead, they improve significantly the AIC score and reduce by almost half the mean confidence interval. In other words,

Conclusions

- **Initial Paradox Explained**: Although uncorrected ARIMA models can occasionally yield lower point-prediction errors due to sheer luck or overfitting to historical noise, they remain statistically invalid.
- **The ARIMAX Advantage**: Incorporating calendar effects and outliers via a SARIMAX(1,1,2) model properly addresses non-stationarity, successfully passes the Ljung-Box test, and eliminates residual patterns.
- **Uncertainty Halved**: By accounting for external variations simultaneously through exogenous regressors ($xreg$), the ARIMAX model slashes the prediction interval width nearly in half, drastically reducing forecasting uncertainty.
- **Strategic Future Projection**: For robust future forecasting, this framework allows the model to seamlessly adapt to upcoming calendar structures while safely neutralizing past anomalous shocks.

Appendices

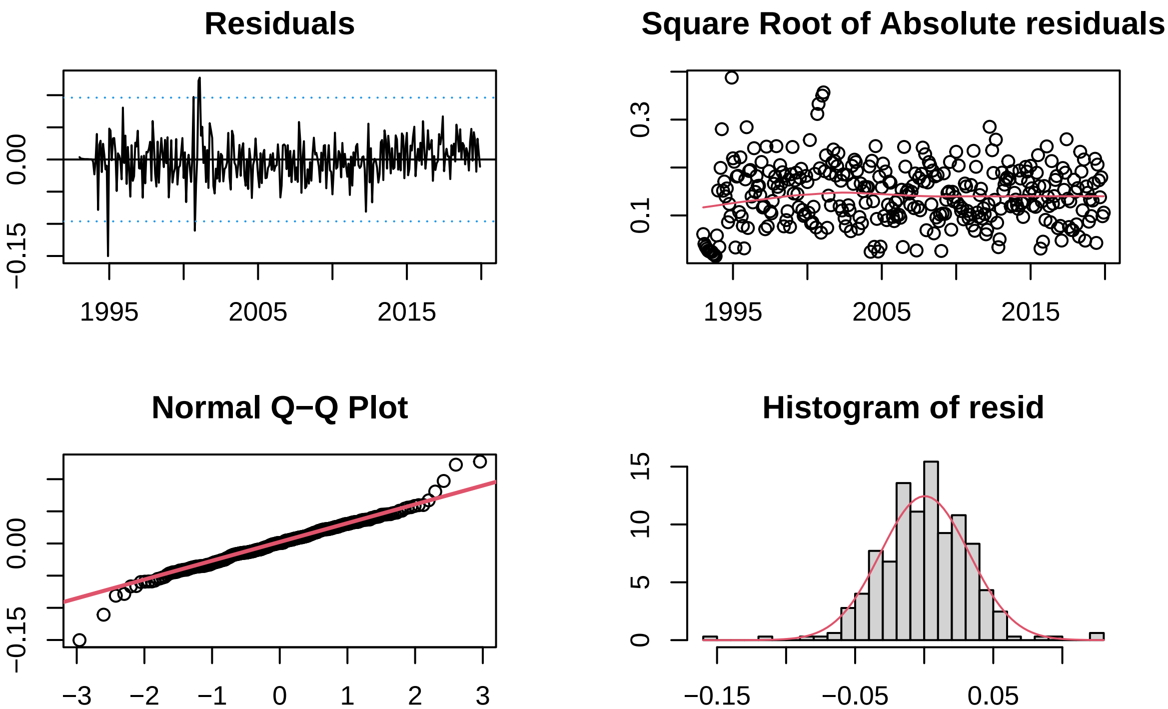
Validation function

```
#################Validation#################################
validation=function(model,dades){
  s=frequency(get(model$series))
  resid=model$residuals
  par(mfrow=c(2,2),mar=c(3,3,3,3))
  #Residuals plot
  plot(resid,main="Residuals")
  abline(h=0)
  abline(h=c(-3*sd(resid),3*sd(resid)),lty=3,col=4)
  #Square Root of absolute values of residuals (Homocedasticity)
  scatter.smooth(sqrt(abs(resid)),main="Square Root of Absolute residuals",
                 lpars=list(col=2))

  #Normal plot of residuals
  qqnorm(resid)
  qqline(resid,col=2,lwd=2)

  ##Histogram of residuals with normal curve
  hist(resid,breaks=20,freq=FALSE)
  curve(dnorm(x,mean=mean(resid),sd=sd(resid)),col=2,add=T)


  #ACF & PACF of residuals
  par(mfrow=c(1,2))
  acf(resid,ylim=c(-1,1),lag.max=60,col=c(2,rep(1,s-1)),lwd=1)
  pacf(resid,ylim=c(-1,1),lag.max=60,col=c(rep(1,s-1),2),lwd=1)
  par(mfrow=c(1,1))

  #ACF & PACF of square residuals
  par(mfrow=c(1,2))
  acf(resid^2,ylim=c(-1,1),lag.max=60,col=c(2,rep(1,s-1)),lwd=1)
  pacf(resid^2,ylim=c(-1,1),lag.max=60,col=c(rep(1,s-1),2),lwd=1)
  par(mfrow=c(1,1))

  #Ljung-Box p-values
  par(mar=c(2,2,1,1))
  tsdiag(model,gof.lag=7*s)
  cat("\n--------------------------------------------------------------------\n")
  print(model)

  #Stationary and Invertible
  cat("\nModul of AR Characteristic polynomial Roots: ",
      Mod(polyroot(c(1,-model$model$phi))),"\n")
  cat("\nModul of MA Characteristic polynomial Roots: ",
      Mod(polyroot(c(1,model$model$theta))),"\n")

  #Model expressed as an MA infinity (psi-weights)
  psis=ARMAtoMA(ar=model$model$phi,ma=model$model$theta,lag.max=36)
  names(psis)=paste("psi",1:36)
  cat("\nPsi-weights (MA(inf))\n")
```

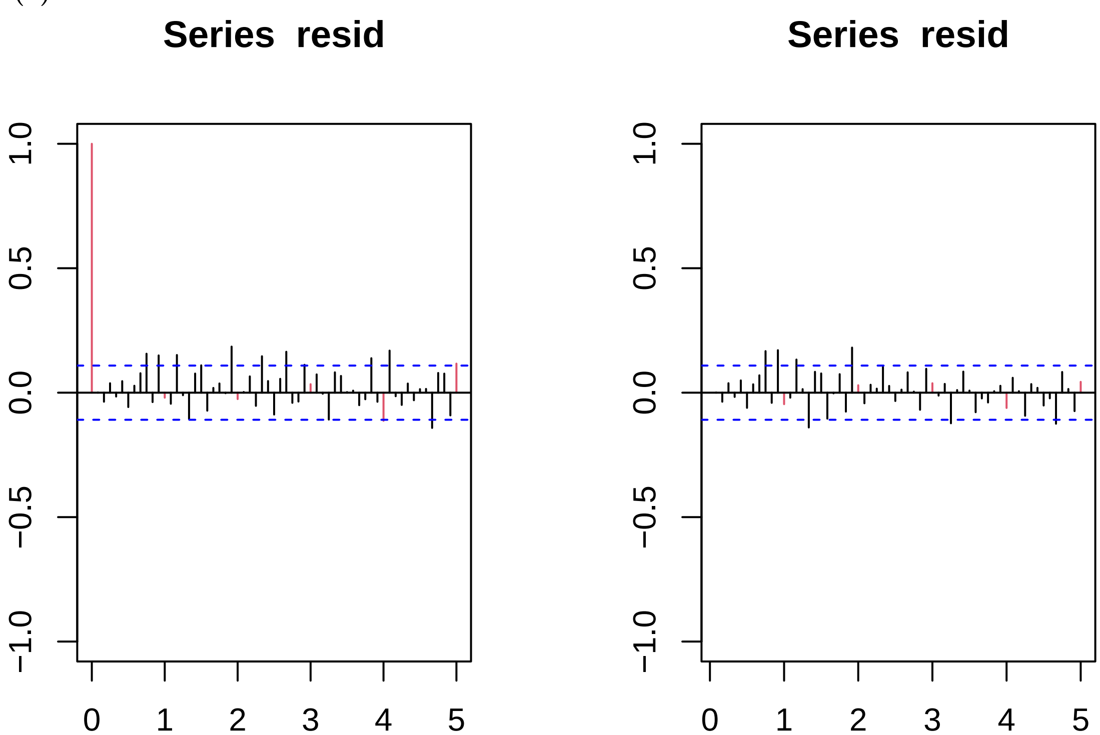
```r
cat("\n--------------------\n")
print(psis[1:20])

#Model expressed as an AR infinity (pi-weights)
pis=-ARMAtoMA(ar=-model$model$theta,ma=-model$model$phi,lag.max=36)
names(pis)=paste("pi",1:36)
cat("\nPi-weights (AR(inf))\n")
cat("\n--------------------\n")
print(pis[1:20])

## Add here complementary tests (use with caution!)
##------------------------------------------------------------
cat("\nNormality Tests\n")
cat("\n--------------------\n")

##Shapiro-Wilks Normality test
print(shapiro.test(resid(model)))

suppressMessages(require(nortest,quietly=TRUE,warn.conflicts=FALSE))
##Anderson-Darling test
print(ad.test(resid(model)))

suppressMessages(require(tseries,quietly=TRUE,warn.conflicts=FALSE))
##Jarque-Bera test
print(jarque.bera.test(resid(model)))

cat("\nHomoscedasticity Test\n")
cat("\n--------------------\n")
suppressMessages(require(lmtest,quietly=TRUE,warn.conflicts=FALSE))
##Breusch-Pagan test
obs=get(model$series)
print(bptest(resid(model)~I(obs-resid(model))))

cat("\nIndependence Tests\n")
cat("\n--------------------\n")

##Durbin-Watson test
print(dwtest(resid(model)~I(1:length(resid(model)))))

##Ljung-Box test
cat("\nLjung-Box test\n")
print(t(apply(matrix(c(1:4,(1:4)*s)),1,function(el) {
  te=Box.test(resid(model),type="Ljung-Box",lag=el)
  c(lag=(te$parameter),statistic=te$statistic[[1]],p.value=te$p.value)})))


#Sample ACF vs. Teoric ACF
par(mfrow=c(2,2),mar=c(3,3,3,3))
acf(dades, ylim=c(-1,1) ,lag.max=36,main="Sample ACF")

plot(ARMAacf(model$model$phi,model$model$theta,lag.max=36),ylim=c(-1,1),
     type="h",xlab="Lag",  ylab="", main="ACF Teoric")
abline(h=0)
```

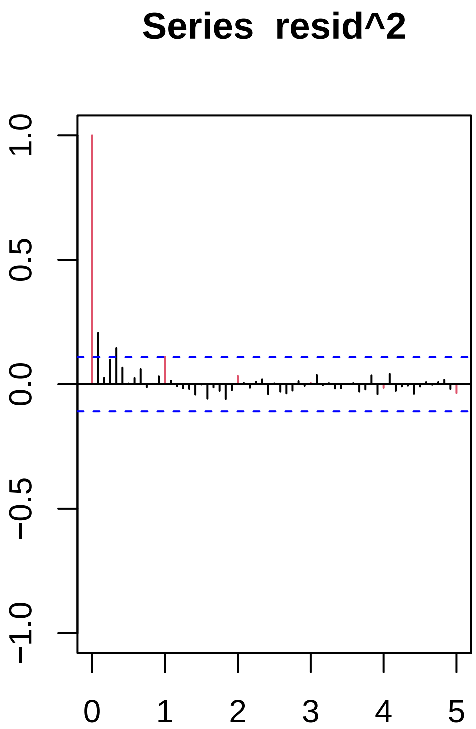
```
  #Sample PACF vs. Teoric PACF
  pacf(dades, ylim=c(-1,1) ,lag.max=36,main="Sample PACF")

  plot(ARMAacf(model$model$phi,model$model$theta,lag.max=36, pacf=T),ylim=c(-1,1),
       type="h", xlab="Lag", ylab="", main="PACF Teoric")
  abline(h=0)
  par(mfrow=c(1,1))
}
################# Fi Validation ##################################
```

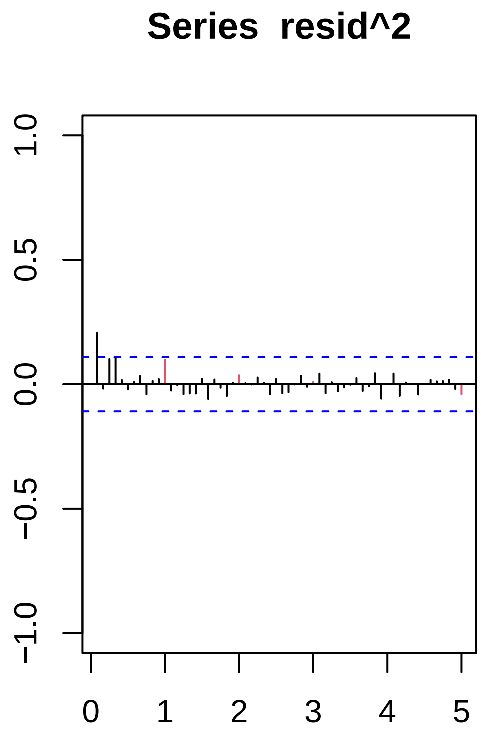
ARIMA models validation detailed

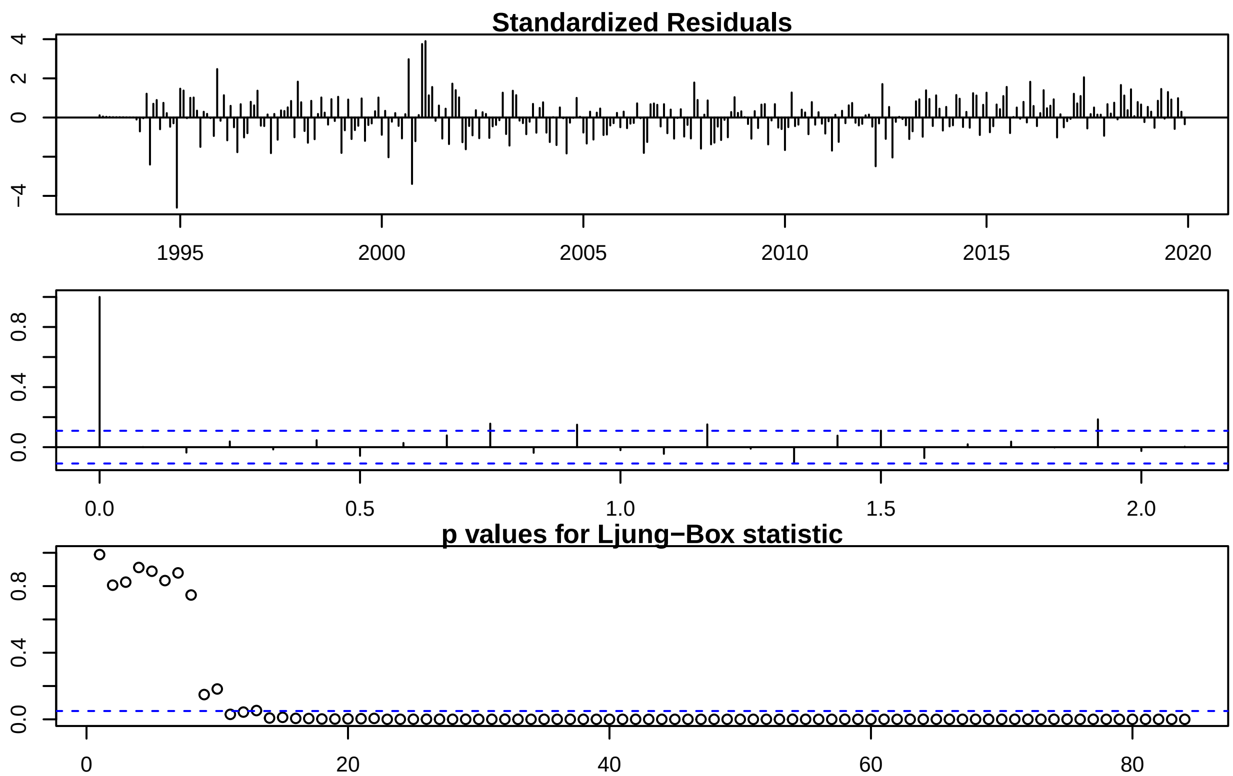


MA(2)





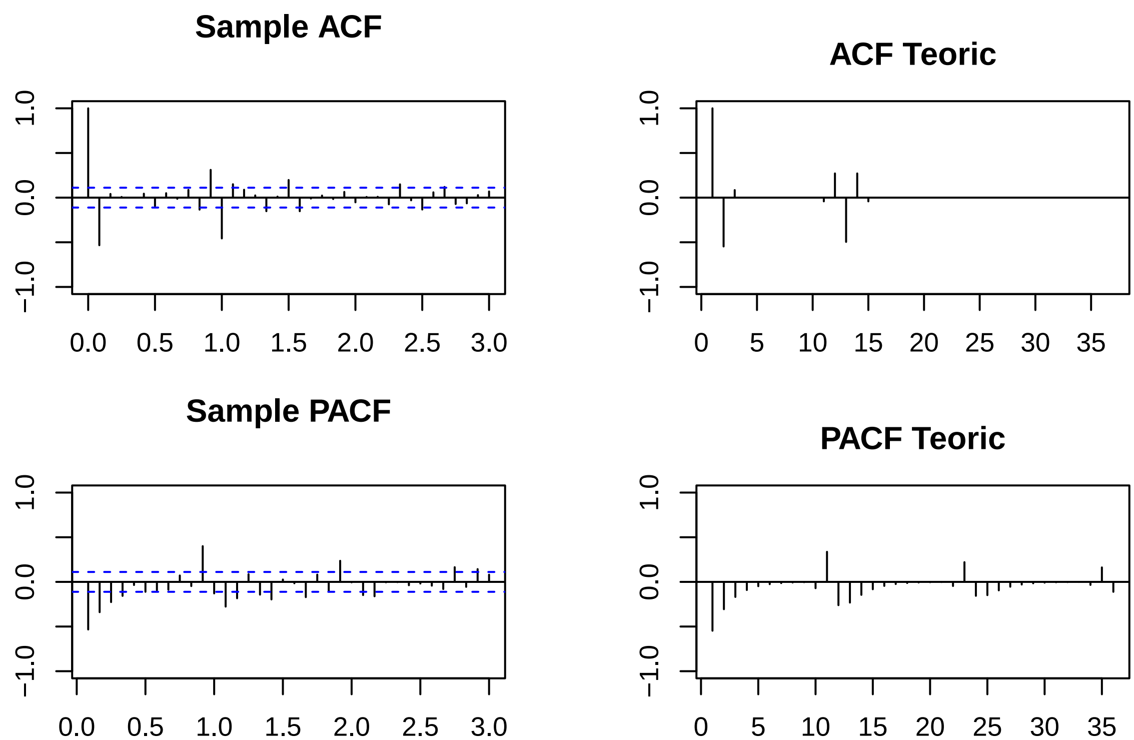


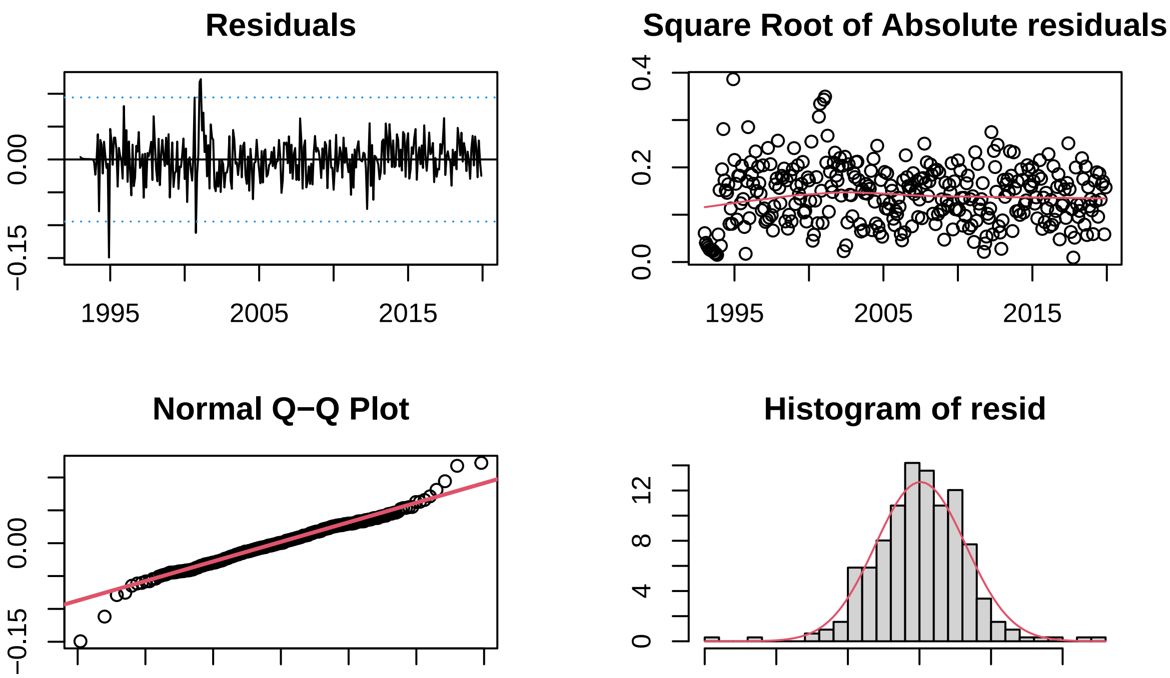



```
##
## --------------------------------------------------------------------
##
## Call:
## arima(x = lnserie, order = c(0, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12))
```

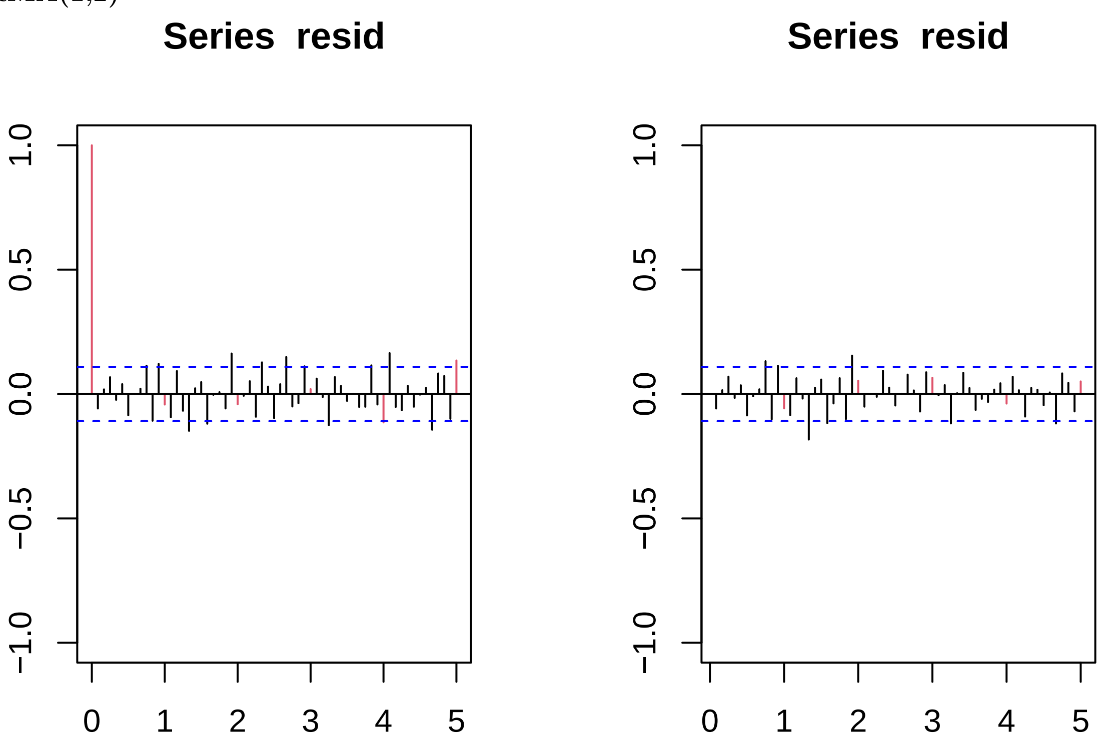
```
##
## Coefficients:
##           ma1     ma2     sma1
##       -0.7872  0.1387  -0.8759
## s.e.   0.0595  0.0556   0.0451
##
## sigma^2 estimated as 0.001065:  log likelihood = 613.95,  aic = -1219.89
##
## Modul of AR Characteristic polynomial Roots:
##
## Modul of MA Characteristic polynomial Roots:  1.0111 1.0111 1.0111 1.0111 1.0111 1.0111 1.0111 1.0111
##
## Psi-weights (MA(inf))
##
## --------------------
##      psi 1      psi 2      psi 3      psi 4      psi 5      psi 6      psi 7
## -0.7872180  0.1386801  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000
##      psi 8      psi 9     psi 10     psi 11     psi 12     psi 13     psi 14
##  0.0000000  0.0000000  0.0000000  0.0000000 -0.8759337  0.6895508 -0.1214746
##     psi 15     psi 16     psi 17     psi 18     psi 19     psi 20
##  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000
##
## Pi-weights (AR(inf))
##
## --------------------
##         pi 1         pi 2         pi 3         pi 4         pi 5         pi 6
## -0.787218034 -0.481032141 -0.269505707 -0.145450171 -0.077125922 -0.040543873
##         pi 7         pi 8         pi 9        pi 10        pi 11        pi 12
## -0.021221038 -0.011082956 -0.005781767 -0.003014526 -0.001571273 -0.876752591
##        pi 13        pi 14        pi 15        pi 16        pi 17        pi 18
## -0.689977547 -0.421574638 -0.236185008 -0.127465088 -0.067588657 -0.035530140
##        pi 19        pi 20
## -0.018596766 -0.009712386
##
## Normality Tests
##
## --------------------
##
##  Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.97599, p-value = 3.076e-05
##
##
##  Anderson-Darling normality test
##
## data:  resid(model)
## A = 0.73796, p-value = 0.05399
##
##
##  Jarque Bera Test
##
## data:  resid(model)
## X-squared = 71.851, df = 2, p-value = 2.22e-16
```

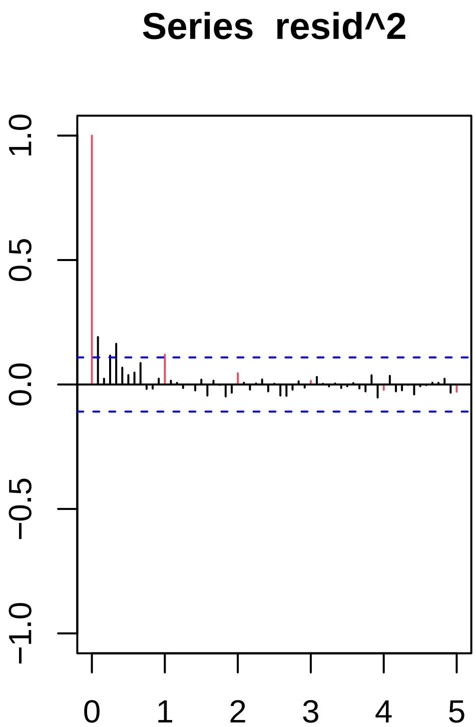
```
##
##
## Homoscedasticity Test
##
## --------------------
##
##  studentized Breusch-Pagan test
##
## data:  resid(model) ~ I(obs - resid(model))
## BP = 2.3425, df = 1, p-value = 0.1259
##
##
## Independence Tests
##
## --------------------
##
##  Durbin-Watson test
##
## data:  resid(model) ~ I(1:length(resid(model)))
## DW = 2.0239, p-value = 0.5635
## alternative hypothesis: true autocorrelation is greater than 0
##
##
## Ljung-Box test
##      lag.df    statistic      p.value
## [1,]      1 1.830573e-04 9.892051e-01
## [2,]      2 4.321965e-01 8.056561e-01
## [3,]      3 9.062973e-01 8.239079e-01
## [4,]      4 9.863685e-01 9.118559e-01
## [5,]     12 2.149488e+01 4.358670e-02
## [6,]     24 5.472970e+01 3.403871e-04
## [7,]     36 8.541745e+01 6.762500e-06
## [8,]     48 1.098332e+02 9.496701e-07
```

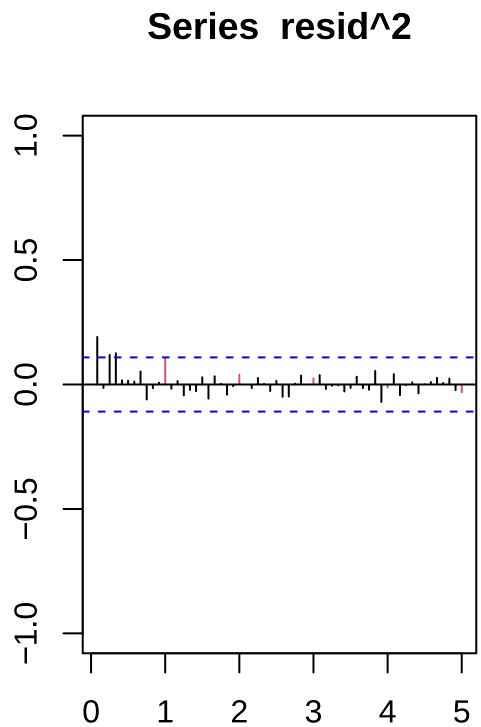

Sample ACF
ACF Teoric
Sample PACF
PACF Teoric
1.0
0.0
−1.0
0.0 0.5 1.0 1.5 2.0 2.5 3.0
0 5 10 15 20 25 30 35

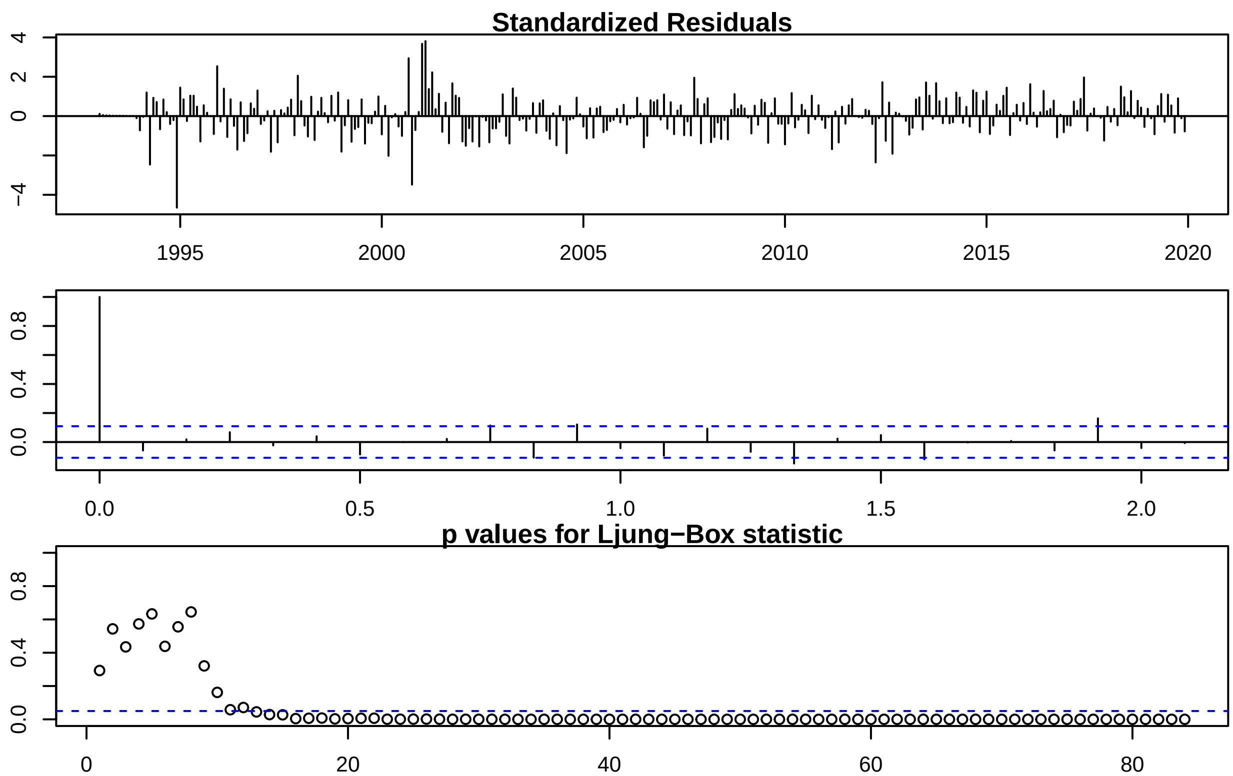

Residuals
Square Root of Absolute residuals
Normal Q−Q Plot
Histogram of resid

ARMA(1,2)


Series resid
Series resid



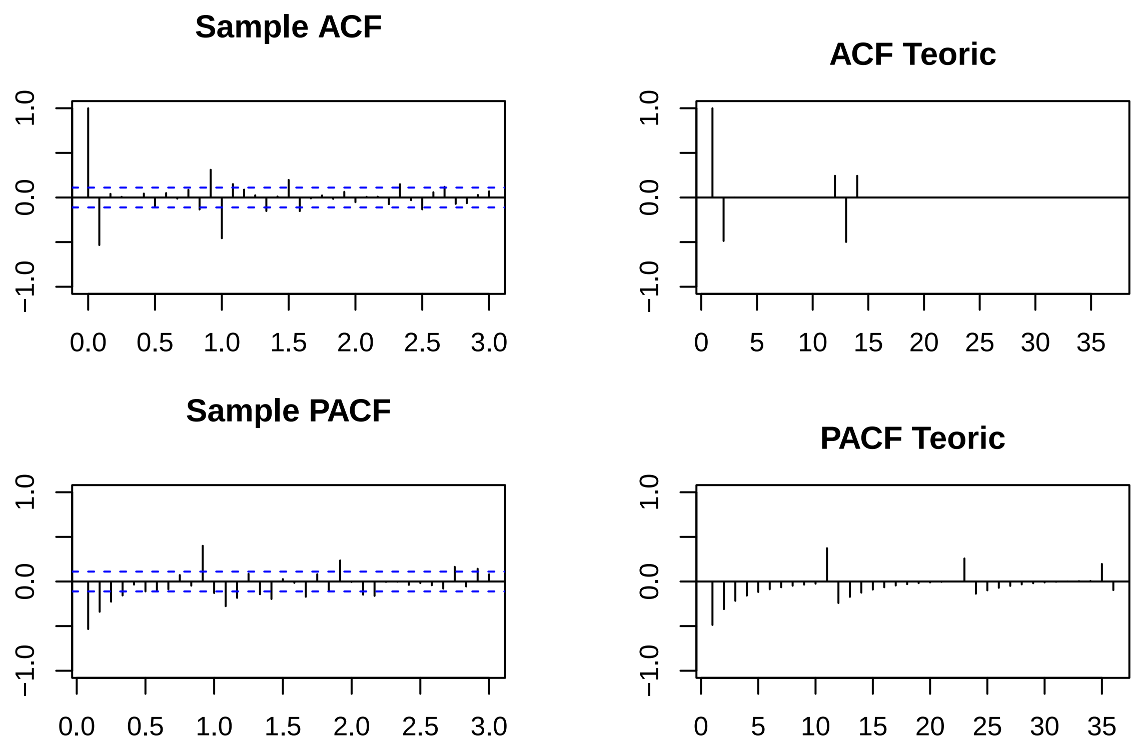





```
##
## ----------------------------------------------------------------------
##
## Call:
## arima(x = lnserie, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12))
```

```
##
## Coefficients:
##          ar1      ma1     ma2     sma1
##       0.9888  -1.7281  0.7451  -0.9219
## s.e.  0.0159   0.0377  0.0356   0.0452
##
## sigma^2 estimated as 0.001028:  log likelihood = 617.75,  aic = -1225.5
##
## Modul of AR Characteristic polynomial Roots:  1.011333
##
## Modul of MA Characteristic polynomial Roots:  1.006796 1.006796 1.006796 1.006796 1.006796 1.006796 
##
## Psi-weights (MA(inf))
##
## --------------------
##         psi 1         psi 2         psi 3         psi 4         psi 5
## -0.7393437608  0.0140457561  0.0138883529  0.0137327135  0.0135788183
##         psi 6         psi 7         psi 8         psi 9        psi 10
##  0.0134266478  0.0132761825  0.0131274034  0.0129802916  0.0128348284
##        psi 11        psi 12        psi 13        psi 14        psi 15
##  0.0126909953 -0.9093917674  0.6940391340 -0.0006802568 -0.0006726335
##        psi 16        psi 17        psi 18        psi 19        psi 20
## -0.0006650957 -0.0006576423 -0.0006502724 -0.0006429852 -0.0006357796
##
## Pi-weights (AR(inf))
##
## --------------------
##         pi 1         pi 2         pi 3         pi 4         pi 5         pi 6
## -0.739343761 -0.532583440 -0.369489249 -0.241698053 -0.142379471 -0.065961066
##         pi 7         pi 8         pi 9        pi 10        pi 11        pi 12
## -0.007902253  0.035491682  0.067222500  0.089724712  0.104968862 -0.807394186
##        pi 13        pi 14        pi 15        pi 16        pi 17        pi 18
## -0.561891887 -0.369433617 -0.219764071 -0.104515990 -0.016870873  0.048720107
##        pi 19        pi 20
##  0.096765590  0.130922674
##
## Normality Tests
##
## --------------------
##
##  Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.9754, p-value = 2.419e-05
##
##
##  Anderson-Darling normality test
##
## data:  resid(model)
## A = 0.81635, p-value = 0.03455
##
##
##  Jarque Bera Test
##
```

```
## data:  resid(model)
## X-squared = 73.466, df = 2, p-value < 2.2e-16
##
##
## Homoscedasticity Test
##
## --------------------
##
##  studentized Breusch-Pagan test
##
## data:  resid(model) ~ I(obs - resid(model))
## BP = 3.1928, df = 1, p-value = 0.07397
##
##
## Independence Tests
##
## --------------------
##
##  Durbin-Watson test
##
## data:  resid(model) ~ I(1:length(resid(model)))
## DW = 2.1273, p-value = 0.8629
## alternative hypothesis: true autocorrelation is greater than 0
##
##
## Ljung-Box test
##      lag.df  statistic      p.value
## [1,]      1   1.105301 2.931057e-01
## [2,]      2   1.220740 5.431499e-01
## [3,]      3   2.730671 4.350402e-01
## [4,]      4   2.910458 5.729201e-01
## [5,]     12  19.796407 7.103661e-02
## [6,]     24  51.747428 8.439303e-04
## [7,]     36  79.893649 3.581262e-05
## [8,]     48 102.282063 8.457064e-06
```


Sample ACF
ACF Teoric
Sample PACF
PACF Teoric
1.0
0.0
−1.0
0.0 0.5 1.0 1.5 2.0 2.5 3.0
0 5 10 15 20 25 30 35